# Asymmetric Bonding and Buyer Dominance: Two Composing Mechanisms for Self-Enforcing $N$-Party Coordination

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**Abstract**

We present FIGARO, a settlement mechanism that enables self-enforcing agreements between mutually distrusting parties without arbitrators, timeouts, or governance. The mechanism composes two distinct primitives. *Asymmetric bonding*: each party locks collateral equal to twice the transaction value, making cooperation the weakly dominant strategy in a one-shot game and the unique profile surviving iterated elimination of weakly dominated strategies. *Buyer dominance with atomic resolution*: only the buyer can trigger resolution, and resolution settles all orders in a process simultaneously or not at all. The two mechanisms are inseparable: bonding alone gives independently secured bilateral edges that cannot multi-party coordinate; buyer dominance alone is without force absent the bonding equilibrium.

Asymmetric bonding scales from two-party exchange to $N$-party process trees through *progressive collateralization*: each seller bonds against the cumulative value of all upstream work, preserving the Nash equilibrium at every chain position with the risk-to-reward ratio growing with chain depth. Buyer dominance with atomic resolution then operates on the already-scaled mesh: the all-or-nothing resolution rule induces a weakest-link subgame among sellers whose peer-enforcement properties we show to obtain under strictly weaker assumptions than the joint-liability mechanisms classically analyzed in the microfinance literature.

This paper is the mechanism-design half of a multi-paper development. The implementation as a Solidity smart contract, with formal verification across multiple methods, is presented in the implementation paper. Implications for organizational economics—specifically, the Coasean shift toward transaction-scoped institutions—are developed in the institutional-economics paper.

**Keywords:** mechanism design, Nash equilibrium, collateral bonding, multi-party coordination, process trees, peer enforcement

## 1 Introduction

### 1.1 The Problem

Economic coordination between strangers requires trust infrastructure. When two parties who have never met wish to exchange value, they face the classic commitment problem: either party may defect after the other has performed. The historical solution is *institutional intermediation*—firms, courts, escrow services, platform operators—each of which introduces costs, delays, jurisdictional dependencies, and power asymmetries.

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The problem intensifies in multi-party settings. A food delivery involves a cook, a kitchen operator, an ingredient sourcer, and a courier; a procurement process involves engineers, manufacturers, inspectors, and logistics providers. The standard architectural response is to aggregate participants into a single legal entity that internalizes the coordination problem and resolves it through hierarchical management.

Blockchain-based smart contracts have produced a rich literature on trustless exchange, but most protocols address financial operations—swaps, lending, derivatives—rather than general coordination. Those that target coordination (e.g., Kleros 2019, Aragon 2017) typically reintroduce the intermediary as an on-chain dispute resolution mechanism: a jury, a governance body, or a timeout that unilaterally releases funds.

### 1.2 Our Contribution

We present FIGARO, which achieves self-enforcing coordination through two composing mechanisms: *asymmetric bonding*, which produces the bilateral Nash equilibrium and scales it to $N$-party process trees via progressive collateralization; and *buyer dominance with atomic resolution*, which enforces inter-seller coordination on the already-scaled mesh. The two mechanisms are inseparable: bonding alone yields independently secured edges that cannot multi-party coordinate; buyer dominance alone is worthless without the bonding equilibrium. Together they make the mesh resolvable from a single signature with cooperation pressure propagating through it. The contributions are:

(i) **A minimal settlement primitive** with no timeout, no governance, and no escape hatches from the bonded state. We show that this minimality is not a simplification but a *security requirement*: every additional exit path weakens the Nash equilibrium (Theorem 4.7).

(ii) **Progressive collateralization** for $N$-party process trees. We prove (Theorem 5.3) that cooperation remains weakly dominant at every position in an arbitrary-depth process tree (with strict dominance whenever at least one other player's cooperation is not already foreclosed), and is the unique profile surviving iterated elimination of weakly dominated strategies; the seller's risk-to-reward ratio increases with chain depth.

(iii) **Atomic resolution** as a coordination forcing function. We show that all-or-nothing process settlement induces a weakest-link subgame among sellers, creating endogenous peer pressure without explicit communication or governance.

(iv) **A reduction over joint-liability microfinance.** We prove (Theorem 6.1) that the peer-enforcement outcome common to Ghatak [1999] and Stiglitz [1990] is obtained by the bonded commitment mechanism under strictly weaker assumptions: no repeated interaction, no local information, no exogenous punishment technology, and no joint-liability contracting.

(v) **Self-enforcing DAG topology** without on-chain validation. We prove (Theorem 5.8) that misreporting cumulative value is a strictly dominated strategy whenever the probability of non-resolution is positive, making on-chain parent-relationship validation unnecessary.

The mechanism's implementation as a Solidity smart contract, the formal-verification stack used to audit it, the threat model, and the three-layer enforcement architecture (economic / coordination / evidence) are developed in the companion implementation paper. The implications of a fixed-cost enforcement primitive for the boundary of the firm in the sense of Coase [1937]—and specifically the demotion of the firm from privileged organizational container to one pattern among many when the subordination overlay is no longer enforcement-driven—are developed in the companion institutional-economics paper.

## 1.3 Paper Organization

Section 2 surveys related work. Section 3 presents the mechanism formally. Section 4 proves the game-theoretic properties. Section 5 extends the model to $N$-party process trees and DAGs. Section 6 states the reduction over joint-liability microfinance. Section 7 concludes with open theoretical questions.

## 1.4 Three Tiers: Kernel, Protocol, Runtime

FIGARO as described in this paper is a mathematical object: a set of definitions, an equilibrium, and the theorems connecting them. Across the companion papers, three tiers are distinguished that the present paper uses only once but that subsequent companions invoke frequently:

- **Kernel** — the irreducible settlement primitive formalized here: the `commit` and `resolveProcess` operations, the bond structure, and the equilibrium they induce. Every claim in the present paper is a claim at this tier.
- **Protocol** — the kernel together with ancillary mechanisms that compose with it under a preservation discipline (attestation coordinators, schema registries, auction primitives, operator registries). The composition discipline itself is developed in the companion implementation paper.
- **Runtime** — the protocol together with semantic layers, builder surfaces, user interfaces, and the vertical-specific deployments that instantiate it for a given industry or community. Runtime choices (specific tokens, specific schemas, specific operator communities) are out-of-scope for the present paper.

Claims about "the primitive" that appear in companion papers refer, unless noted otherwise, to the kernel tier. Properties sometimes loosely attributed to the primitive — schema-typed attestations, NFT-anchored credentials, specific token denominations, accepted-token lists — live at the protocol or runtime tiers and are guaranteed only by the additional mechanisms that implement them.

**The kernel is ideologically agnostic; the graph is the politics.** A consequence of locating the whole of the present paper's argument at the kernel tier is that the kernel admits no commitments about *what* is being coordinated or *under what normative regime*. Currency, jurisdiction, identity type, arbitration forum, role structure, price-discovery mechanism, contribution metric — none of these are in the kernel; all of them are expressed in the *graph* that parties compose on top of it. A Dutch-auction-and-private-operator graph expresses a market-liberal coordination pattern; a cooperative-bonding-and-contribution-weighted-resolution graph expresses a cooperative one; an interest-free risk-sharing graph expresses an Islamic-finance one. The kernel enables each and prefers none. This is not evasion. It is the architectural consequence of placing the enforcement function at TCP/IP altitude: a layer below which nothing remains to factor out, and a layer above which the normative choices become legible as choices rather than as properties of the infrastructure. Claims in the companion papers that bear on ideology or normative regime — the political-economy companion paper on cultural hegemony; the labor-law companion paper on subordination — are claims about what becomes *expressible* in the graph once the kernel is in place, not claims about the kernel itself.

**Composability is external-first.** A second consequence of the kernel's narrow construction is that the protocol's useful extensions are predominantly *external* rather than internal. The kernel does not include a dispute forum (Kleros, the Singapore International Arbitration Centre, or any court); it does not include a carbon-offset market (Klima DAO, Toucan, Moss); it does not include a prediction market, an insurance pool, a lending facility, a tax-reporting service, an identity provider, a storage layer, or a messaging fabric. Each of these exists as a working external protocol or institution, and an assembly composed at the runtime tier can name any of them in its graph: a forum-clause names the dispute forum the parties agree to; an attestation clause names the offset operator the buyer wants to compose into the process tree; a sub-order edge in the tree carries the buyer's offset purchase to the named operator within the same atomic resolution as the original commerce. Modern wallets handle the multi-token bookkeeping that this composition implies, which is why the kernel itself does not need to (and, by Theorem 4.7, must not). The composability posture is the architectural counterpart of the ideological-agnosticism claim above: the kernel makes the external surfaces composable; the graph names which externals it composes with. The companion implementation paper (§7) develops the formal *coordinator pattern* that makes such composition equilibrium-preserving; the present paper records only the architectural posture from which the pattern follows.

# 2 Related Work

**Mutual-destruction escrows.** The direct ancestor of FIGARO is the Safe Remote Purchase contract [Solidity Team, 2016], included in the Solidity documentation as a reference pattern: buyer and seller each lock 2× the item value, and only mutual agreement releases the funds. This two-party, single-order pattern is the seed from which FIGARO grew. BitHalo [BitHalo, 2014] independently explored mutual-destruction deposits for peer-to-peer Bitcoin trade.

The central problem was generalizing SRP beyond two parties. SRP uses symmetric 2× deposits with mutual agreement to release—both parties must cooperate. This works for a single exchange but cannot compose: in a twelve-seller delivery workflow, there is no natural "mutual agreement" among all participants. *Asymmetric bonding* (Theorem 3.1) solves this. By scaling each seller's bond with cumulative upstream value rather than fixing it per-trade, the mechanism creates a mesh of independently secured edges. Each edge carries its own Nash equilibrium (Theorem 4.3), and the mesh decomposes as finely as is operationally tolerable.

*Buyer dominance* then takes advantage of this mesh structure to make multi-party coordination resolvable from a single signature. Because a single party resolves the entire process tree atomically, coordination pressure propagates through the mesh without any management layer (Theorem 5.11).

**State channels and payment networks.** The Lightning Network [Poon and Dryja, 2016] and Raiden [Raiden, 2017] use time-locked deposits for off-chain payment. These protocols optimize for high-frequency bilateral payments and rely on timeouts for dispute resolution—precisely the escape hatch that FIGARO eliminates (Theorem 4.7). More critically, state channels do not address multi-party *coordination* (delivery, fulfillment, service composition); they address multi-hop *payment routing.*

**On-chain dispute resolution.** Kleros [Kleros, 2019] provides decentralized arbitration via Schelling-point juror selection. Aragon Court [Aragon, 2017] offers similar functionality within DAO governance. Both reintroduce a *discretionary human decision* into the resolution path, breaking the self-enforcing property (Theorem 4.7, Case 2).

**Optimistic mechanisms.** Optimistic rollups [Optimism, 2021] and optimistic oracles [UMA, 2020] use a challenge-response pattern: an assertion stands unless challenged within a timeout window. This requires liveness assumptions (the challenger must be online) and timeout calibration (too short enables fraud; too long delays finality). FIGARO has no timeout from the bonded state—capital remains locked indefinitely until the buyer resolves, making liveness irrelevant to security.

**Mechanism design.** The revelation principle [Myerson, 1979] guarantees that any achievable outcome can be implemented via a direct mechanism. FIGARO's insight is that the "direct mechanism" for multi-party coordination need not involve message passing, reporting, or arbitration at all: locked capital *is* the mechanism. The closest theoretical antecedent is the concept of *deposit games* [Asgaonkar and Krishnamachari, 2019], which analyze two-party security deposits. We extend deposit-game analysis to $N$-party DAGs with progressive collateralization.

**Joint-liability microfinance.** Group lending models—most prominently the Grameen Bank [Yunus, 1999] and its formalizations by Ghatak [1999] and Stiglitz [1990]—show that joint liability combined with a social substrate can produce peer selection and peer monitoring outcomes. We return to these models in Section 6: FIGARO reproduces the same peer-enforcement outcome under strictly weaker assumptions, with no appeal to repeated interaction or local information.

# 3 The Two Mechanisms: Formal Definition

## 3.1 Notation and Setup

Consider two parties, a *buyer* $B$ and a *seller* $S$, who wish to exchange a payment $P > 0$ (denominated in a unit of account) for goods or services. Both parties hold sufficient balances.

**Definition 3.1** (Asymmetric Bonding)**.** An *asymmetric bond* for a transaction with payment $P$ and cumulative value $G \geq P$ is a pair of deposits:

$$C_b = 2P \quad \text{(buyer bond)}, \tag{1}$$

$$C_s = 2G \quad \text{(seller bond)}. \tag{2}$$

For a root transaction (no upstream value), $G = P$, so both bonds equal $2P$. For sub-orders in a process tree, $G > P$, creating an asymmetry where the seller's bond exceeds the buyer's.

**Definition 3.2** (Commitment)**.** A *commitment* is a tuple $\langle B, S, \tau, P, h, \sigma, \delta \rangle$ where $B$ is the buyer identity, $S$ the seller identity, $\tau$ the unit of account, $P$ the payment, $h$ a content hash of the off-chain agreement, $\sigma$ a cryptographic salt, and $\delta$ a signature expiry deadline. Both parties sign the commitment off-chain.

**Definition 3.3** (Process)**.** A *process* $\Pi$ is a tuple $\langle \text{id}, B, \tau, G, n, \mathcal{O} \rangle$ where id is a deterministic identifier derived from the root commitment, $B$ is the root buyer (constant across all orders), $\tau$ is the denomination (constant), $G$ is a *monotonic accumulator* tracking total committed value, $n$ is the count of active orders, and $\mathcal{O}$ is the set of active order identifiers.

## 3.2 Protocol Operations

The mechanism exposes exactly two state-changing operations.

**Operation 1: `commit`.** Given a commitment $c$ signed by both $B$ and $S$:

1. Verify both signatures.
2. If $c$.processId $= 0$, derive a new process identifier from the commitment digest and initialize the process with $G \leftarrow P$ and $n \leftarrow 1$.
3. Otherwise verify that $c$.processId identifies an existing process, that signer $B$ matches the process root buyer (*buyer dominance*), and that $c$.expectedCumulativeValue $= \Pi.G + P$.
4. Pull $C_b = 2P$ from $B$ and $C_s = 2c$.expectedCumulativeValue from $S$.
5. Create or extend the process accordingly, update the monotonic accumulator, and emit an evidence event that fully records the commitment.

**Operation 2:** `resolveProcess`**.** Given a process identifier and the complete list of active commitments:

1. Verify the caller identity equals $\Pi.B$ (buyer dominance).
2. Verify that the list length equals $\Pi.n$ (atomic resolution: *all* orders must be included).
3. For each order $o_i$ with payment $P_i$ and cumulative snapshot $G_i$:

$$\pi_{s,i} = 2G_i + P_i, \tag{3}$$
$$\pi_{b,i} = P_i. \tag{4}$$

4. Transfer $\pi_{s,i}$ to each seller, $\pi_{b,i}$ to the buyer.
5. Mark all orders as resolved.

*Remark* 3.4 (Conservation Law)*.* For each order, the total distributed equals total locked:

$$\pi_{s,i} + \pi_{b,i} = (2G_i + P_i) + P_i = 2G_i + 2P_i = C_{s,i} + C_{b,i}. \tag{5}$$

The escrow holds zero balance after resolution.

*Remark* 3.5 (No Escape Hatches)*.* There is no timeout, no cancel-after-commitment, no admin override, no governance vote, and no unilateral exit from the bonded state. Once both parties have committed, the only path to fund recovery is buyer-initiated resolution. This is not a temporary simplification; it is a *security requirement.* Theorem 4.7 proves that any additional exit path weakens the Nash equilibrium.

### 3.3 Net Payoffs

**Definition 3.6** (Generalized Payout Functions)**.** For a bond multiplier $k \geq 1$, an order with payment $P > 0$ and cumulative value $G \geq P$, define the *settlement payout functions*:

$$\pi_s(k, G, P) = k \cdot G + P, \tag{6}$$
$$\pi_b(k, P) = (k - 1) \cdot P. \tag{7}$$

These satisfy the conservation law $\pi_s + \pi_b = k \cdot G + k \cdot P = C_s + C_b$ for $C_s = k \cdot G$ and $C_b = k \cdot P$. The *net payoff* is the payout minus the deposited bond:

$$u_S^{\text{net}}(k) = \pi_s - C_s = P, \tag{8}$$
$$u_B^{\text{net}}(k) = \pi_b - C_b = -P. \tag{9}$$

*Remark* 3.7*.* For the mechanism's $k = 2$: $\pi_s = 2G + P$, $\pi_b = P$, $C_s = 2G$, $C_b = 2P$. The net payoffs ($+P$ for the seller, $-P$ for the buyer) are independent of $k$—the multiplier affects only the capital at risk, not the final value transfer.

# 4 Game-Theoretic Analysis

## 4.1 Two-Party Nash Equilibrium

**Definition 4.1** (The Bonding Game)**.** The *bonding game* $\Gamma = (\{B, S\}, \{A_B, A_S\}, \{u_B, u_S\})$ is a one-shot, simultaneous-move, complete-information game:

- **Players**: buyer $B$, seller $S$.
- **Action sets**: $A_B = A_S = \{\mathrm{C}, \mathrm{D}\}$.

- **Interpretation**: C for $S$ means delivering the agreed goods/services; C for $B$ means calling `resolveProcess`. D means failing to perform.
- **Payoff functions** (net of bond deposits), for a root order with payment $P > 0$ and $G = P$:

$$u_B(a_B, a_S) = \begin{cases} -P & \text{if } a_B = \mathrm{C} \text{ and } a_S = \mathrm{C}, \\ -2P & \text{otherwise.} \end{cases} \tag{10}$$

$$u_S(a_B, a_S) = \begin{cases} +P & \text{if } a_B = \mathrm{C} \text{ and } a_S = \mathrm{C}, \\ -2G & \text{otherwise.} \end{cases} \tag{11}$$

The "otherwise" payoff is identical across all non-cooperative outcomes because any defection prevents resolution: bonds remain locked indefinitely, yielding a net loss equal to the full deposit.

Table 1: Payoff matrix for the bonding game $\Gamma$. Entry format: $(u_B, u_S)$. Parameters: $P > 0$, $G \geq P > 0$.

| | $a_B = \mathrm{C}$ | $a_B = \mathrm{D}$ |
|---|---|---|
| $a_S = \mathrm{C}$ | $(-P,\ +P)$ | $(-2P,\ -2G)$ |
| $a_S = \mathrm{D}$ | $(-2P,\ -2G)$ | $(-2P,\ -2G)$ |

**Definition 4.2** (Dominance)**.** Action $a_i$ *weakly dominates* $a'_i$ for player $i$ if $u_i(a_i, a_{-i}) \geq u_i(a'_i, a_{-i})$ for all $a_{-i}$, with strict inequality for at least one $a_{-i}$. If the inequality is strict for *all* $a_{-i}$, the dominance is *strict*.

**Theorem 4.3** (Two-Party Nash Equilibrium)**.** *In the bonding game* $\Gamma$ *of Theorem 4.1:*

*(a)* C *weakly dominates* D *for both the buyer and the seller.*

*(b)* $(\mathrm{C}, \mathrm{C})$ *is the unique profile surviving iterated elimination of weakly dominated strategies. The profile* $(\mathrm{D}, \mathrm{D})$ *also satisfies the Nash condition (no* profitable *deviation, since both yield the locked-bond payoff), but is weakly dominated and is removed in the first elimination round; we adopt the IEWDS refinement, standard for mechanism design [Myerson, 1979], as the operative equilibrium concept.*

*(c)* $(\mathrm{C}, \mathrm{C})$ *is the unique Pareto-optimal outcome.*

*Proof.* **Part (a): Weak dominance.**

*Buyer.* We compare $u_B(\mathrm{C}, a_S)$ vs. $u_B(\mathrm{D}, a_S)$ for each seller action:

$$a_S = \mathrm{C}: \quad u_B(\mathrm{C}, \mathrm{C}) = -P > -2P = u_B(\mathrm{D}, \mathrm{C}), \tag{12}$$

$$a_S = \mathrm{D}: \quad u_B(\mathrm{C}, \mathrm{D}) = -2P = u_B(\mathrm{D}, \mathrm{D}). \tag{13}$$

Inequality (12) is strict (since $P > 0$); (13) is an equality. By Theorem 4.2, C weakly dominates D for $B$.

*Seller.* We compare $u_S(a_B, \mathrm{C})$ vs. $u_S(a_B, \mathrm{D})$ for each buyer action:

$$a_B = \mathrm{C}: \quad u_S(\mathrm{C}, \mathrm{C}) = +P > -2G = u_S(\mathrm{C}, \mathrm{D}), \tag{14}$$

$$a_B = \mathrm{D}: \quad u_S(\mathrm{D}, \mathrm{C}) = -2G = u_S(\mathrm{D}, \mathrm{D}). \tag{15}$$

Inequality (14) is strict (since $P > 0$ and $G > 0$ imply $P + 2G > 0$); (15) is an equality. C weakly dominates D for $S$.

**Part (b): Unique Nash equilibrium.** Since C weakly dominates D for both players, neither can profitably deviate from $(\mathrm{C}, \mathrm{C})$. We verify no other profile is a Nash equilibrium:

- $(\mathrm{C}, \mathrm{D})$: Seller deviates to C, gaining $P - (-2G) = P + 2G > 0$. Not a NE.
- $(\mathrm{D}, \mathrm{C})$: Buyer deviates to C, gaining $-P - (-2P) = P > 0$. Not a NE.
- $(\mathrm{D}, \mathrm{D})$: Buyer deviates to C with no change (both yield $-2P$). Seller deviates to C with no change (both yield $-2G$). This profile satisfies the Nash condition (no *profitable* deviation), but an iterated elimination of weakly dominated strategies removes D for both players, leaving $(\mathrm{C}, \mathrm{C})$ as the unique survivor.

**Part (c): Pareto optimality.** The payoff vector $(-P, +P)$ Pareto-dominates $(-2P, -2G)$ since $-P > -2P$ and $+P > -2G$. □

*Remark* 4.4 (On Weak vs. Strict Dominance). The dominance is weak, not strict, because when the counterparty defects, both actions yield the same locked-capital payoff. This is intrinsic to the mechanism: once one party defects, the outcome is fully determined regardless of the other's choice. The critical property is that $(\mathrm{D}, \mathrm{D})$ does not survive iterated elimination of weakly dominated strategies—the standard refinement for mechanism design [Myerson, 1979].

### 4.2 Sufficiency of the $2\times$ Multiplier

We now characterize the multiplier $k$ in the generalized bonding game where $C_s = k{\cdot}G$, $C_b = k{\cdot}P$, and payouts are given by Theorem 3.6.

**Lemma 4.5** (Buyer Indifference at $k = 1$)**.** *For $k = 1$, the buyer is indifferent between cooperating and defecting when the seller cooperates: $u_B(\mathrm{C}, \mathrm{C}) = u_B(\mathrm{D}, \mathrm{C}) = -P$.*

*Proof.* With $k = 1$: $C_b = P$ and $\pi_b = (k-1)P = 0$ (from (7)). Under our accounting convention (Theorem 3.6), the buyer's net payoff under cooperation is $u_B(\mathrm{C}, \mathrm{C}) = \pi_b - C_b = 0 - P = -P$. Under defection, the buyer's bond remains locked: $u_B(\mathrm{D}, \mathrm{C}) = -C_b = -P$. Since $u_B(\mathrm{C}, \mathrm{C}) = u_B(\mathrm{D}, \mathrm{C}) = -P$, the buyer has no strictly profitable deviation from D—the mechanism provides zero marginal incentive to resolve. □

**Theorem 4.6** (Minimal Viable Bond Multiplier)**.** *Let $k \in \mathbb{R}_{\geq 1}$ be the bond multiplier. Then:*

*(a) For $k = 1$,* C *does not weakly dominate* D *for the buyer: the buyer is indifferent when the seller cooperates.*

*(b) For every $k > 1$,* C *weakly dominates* D *for both players (Theorem 4.3); $k = 1$ is therefore the infimum of the multipliers for which weak dominance fails on the buyer side.*

*(c) Among integer multipliers, $k = 2$ is the smallest value at which weak dominance holds. For any $k > 2$, the equilibrium properties are identical to $k = 2$, but capital requirements increase by $(k - 2)(G + P)$ per order with no improvement in dominance relations.*

*Thus $k = 2$ is the minimal* integer *multiplier that yields weak dominance of* C *for both players, and is Pareto-optimal in capital efficiency among all integer multipliers; the underlying real-line cutoff is $k = 1$. The integer restriction reflects the discrete denomination of bonded transfers in the deployed setting, not a property of the equilibrium analysis itself.*

*Proof.* **Part (a).** See Theorem 4.5. With $k = 1$, the buyer recovers $\pi_b = 0$ upon resolution, having deposited $C_b = P$. The net monetary payoff from cooperating is $-P$; the net from defecting is also $-P$ (bond locked). The buyer cannot be motivated by the mechanism alone to resolve.

**Part (b).** For $k > 1$: $\pi_b = (k-1)P$, $C_b = kP$. Cooperation yields $u_B(\mathrm{C}, \mathrm{C}) = (k-1)P - kP = -P$; defection yields $u_B(\mathrm{D}, \mathrm{C}) = -kP$. The cooperation–defection gap is $-P - (-kP) = (k-1)P$, which is strictly positive for every $k > 1$. Weak dominance for the buyer therefore holds for every $k > 1$ and fails at $k = 1$ (the gap collapses to zero), establishing $k = 1$ as the infimum at which weak dominance fails on the buyer side. The seller-side weak dominance follows from Theorem 4.3(a) for any $k \geq 1$ (the seller's net payoff under cooperation is $+P$ regardless of $k$, while defection forfeits the bond $kG$).

**Part (c).** For integer $k$, the smallest value at which both Part (a) (failure at $k = 1$) and Part (b) (success for $k > 1$) admit a feasible bond is $k = 2$. The surplus capital locked at $k > 2$, $(k-2)P$ for the buyer and $(k-2)G$ for the seller, earns zero return and is pure deadweight cost. □

### 4.3 Escape-Hatch Weakness

**Theorem 4.7** (Escape-Hatch Weakness). *Let $\Gamma_E$ be the bonding game augmented with a unilateral exit action $E$ available to player $i$, returning fraction $\alpha \in (0, 1]$ of player $i$'s bond. Then either:*

*(a)* C *no longer weakly dominates* D *for at least one player (the Nash equilibrium of Theorem 4.3 may not survive), or*

*(b) $E$ requires a decision by a third party $J \notin \{B, S\}$ whose incentives are not constrained by the bond structure.*

*Proof.* We analyze three exhaustive sub-cases of unilateral exit.

*Case 1: Timeout (automatic exit after duration $T$).* The action space becomes $A_B = \{\mathrm{C}, \mathrm{D}, \mathrm{Wait}\}$ where Wait means the buyer does not resolve and collects $\alpha \cdot C_b$ after time $T$. The buyer's payoff under Wait against a cooperating seller is:

$$u_B(\mathrm{Wait}, \mathrm{C}) = -C_b + \alpha \cdot C_b = -2P(1-\alpha). \tag{16}$$

Compare with cooperation: $u_B(\mathrm{C}, \mathrm{C}) = -P$. The buyer prefers C to Wait iff:

$$-P > -2P(1-\alpha) \iff 2(1-\alpha) > 1 \iff \alpha < \tfrac{1}{2}. \tag{17}$$

For $\alpha \geq \frac{1}{2}$, Wait yields $u_B \geq -P = u_B(\mathrm{C}, \mathrm{C})$, so C no longer (weakly) dominates Wait. The buyer may rationally wait out the timeout instead of resolving. Even for $\alpha < \frac{1}{2}$, the defection cost drops from $2P$ to $2P(1-\alpha)$, weakening the deterrent. The seller, anticipating the timeout, faces a modified game where the buyer's commitment to resolve is no longer credible—the timeout provides an alternative path to partial capital recovery, undermining the "no escape" property that sustains the equilibrium.

*Case 2: Arbitrator override.* A third party $J$ can invoke $E$ on behalf of either player. Formally, this extends the game to three players $\{B, S, J\}$. The mechanism's self-enforcing property requires that the payoff structure *alone* determines the outcome. Introducing $J$ makes the outcome contingent on $J$'s action, and $J$ is not bonded: $J$ deposits nothing, risks nothing, and has no mechanism-constrained incentive to decide correctly. Whether $J$ is an individual (arbitrator), an algorithm (oracle), or a collective (jury), the resolution path now depends on an agent outside the bond structure. This is condition (b).

*Case 3: Governance vote.* A special case of Case 2 where $J$ is replaced by a set of voters $\{J_1, \ldots, J_m\}$ who collectively decide on $E$. This inherits all problems of Case 2, plus: (i) coordination failures among voters, (ii) plutocratic capture if voting power is token-weighted, (iii) voter apathy (rational ignorance). Each voter $J_k$ is unbonded.

In all three cases, either C ceases to weakly dominate all alternatives (Case 1), or resolution depends on an unbonded external agent (Cases 2–3). □

*Remark* 4.8 (On Judicial Review and the Distinction from Cases 2–3). The theorem's "unbonded external agent" condition targets actors whose decision is *constitutive* of the resolution path internal to the mechanism — arbitrators, juries, oracles, voters embedded in the bonding game itself. It does not target the external legal forum that may, on the evidence record, adjudicate a dispute *outside* the bonding game under doctrines of duress, frustration, impossibility, unconscionability, or public policy. Those forums are constrained by their own institutional bond structures (appellate review, professional sanction, doctrinal consistency) and operate on the bonded commitment as evidentiary input rather than as a resolution path within the mechanism. The distinction is elaborated in the companion evidence-and-law paper.

**Corollary 4.9.** *No timeout duration $T$ and no recovery fraction $\alpha > 0$ can be added to the bonding game without weakening the buyer's incentive to cooperate. In particular, full recovery ($\alpha = 1$) reduces the game to a zero-cost option, making defection strictly dominant for the buyer.*

*Remark* 4.10 (Robustness to Bounded Rationality)*.* The dominance results of this section assume full rationality. Two structural properties suggest the equilibrium is robust to weaker conditions. First, the payoff gap is not marginal: the cost of defection for seller $S_i$ is $2G_i$, which grows with chain depth. The asymmetry between cooperation and defection at the choice margin, rather than at the absolute-magnitude margin, is what loss aversion sharpens: prospect theory predicts diminishing sensitivity *within* the loss domain, but the kink at the reference point (between losing nothing and losing the bond) is amplified by loss aversion. The mechanism's deterrent therefore relies on the kink, not on a comparison of large losses against each other. Second, the dominance argument requires only one comparison per player: does cooperating yield a better outcome than defecting, given each possible counterparty action? This is the minimal reasoning task in any strategic setting. It does not require iterated reasoning about counterparty rationality, common knowledge of payoffs, or beliefs about types. What remains genuinely open is behaviour under non-pecuniary preferences—spite, fairness norms, or intrinsic motivation to defect—where the payoff matrix does not capture all relevant utilities.

# 5 $N$-Party Process Trees

## 5.1 Progressive Collateralization

We now extend the two-party model to an arbitrary-length chain of sellers.

**Definition 5.1** (Process Tree)**.** A *process tree* is a sequence of bonded orders $\langle o_1, o_2, \ldots, o_n \rangle$ sharing a common buyer $B$ and denomination $\tau$. Order $o_i$ has local payment $P_i > 0$ and cumulative value:

$$G_i = \sum_{j=1}^{i} P_j. \tag{18}$$

The seller of $o_i$ posts bond $C_{s,i} = 2G_i$; the buyer posts bond $C_{b,i} = 2P_i$.

**Definition 5.2** ($N$-Party Bonding Game)**.** The *$N$-party bonding game* $\Gamma_N = (\{B, S_1, \ldots, S_n\}, \{A_i\}, \{u_i\})$ extends $\Gamma$ to $n$ sellers. Each seller $S_i$ has action set $\{\mathrm{C}, \mathrm{D}\}$. The buyer has action set $\{\mathrm{C}, \mathrm{D}\}$ and chooses after observing each seller's realized action; we assume perfect monitoring, so $B$ perfectly observes whether each $S_i$ delivered, and plays the conditional strategy "resolve iff all sellers cooperated." The perfect-monitoring assumption corresponds to the deployed setting, in which the buyer receives either the agreed goods or not, and the off-chain observation is prior to the on-chain `resolveProcess` call. When monitoring is imperfect, the conditional strategy must be replaced by a signal-contingent resolution rule; we discuss this extension in Section 7. Payoffs under perfect monitoring:

$$u_{S_i}(a_B, a_{S_1}, \ldots, a_{S_n}) = \begin{cases} +P_i & \text{if } a_B = \mathrm{C} \text{ and } a_{S_j} = \mathrm{C}\ \forall j, \\ -2G_i & \text{otherwise.} \end{cases} \tag{19}$$

The buyer's payoff is:

$$u_B(a_B, a_{S_1}, \ldots, a_{S_n}) = \begin{cases} -\sum_{i=1}^{n} P_i & \text{if } a_B = \mathrm{C} \text{ and } a_{S_j} = \mathrm{C}\ \forall j, \\ -\sum_{i=1}^{n} 2P_i & \text{otherwise.} \end{cases} \tag{20}$$

**Theorem 5.3** ($N$-Party Nash Equilibrium)**.** *In $\Gamma_N$, cooperation is a weakly dominant strategy for every seller $S_i$ at every position $i \in \{1, \ldots, n\}$, and $(\mathrm{C}, \mathrm{C}, \ldots, \mathrm{C})$ is the unique outcome surviving iterated elimination of weakly dominated strategies.*

*Proof.* Fix an arbitrary seller $S_i$. Consider the two cases for the remaining players' joint action profile $a_{-i}$.

*Case 1: All other sellers cooperate and buyer resolves.* Then:

$$u_{S_i}(\ldots, \mathrm{C}_i, \ldots) = +P_i, \tag{21}$$
$$u_{S_i}(\ldots, \mathrm{D}_i, \ldots) = -2G_i. \tag{22}$$

Since $P_i > 0$ and $G_i > 0$, we have $P_i > -2G_i$, i.e., $P_i + 2G_i > 0$. The cooperation payoff strictly exceeds the defection payoff.

*Case 2: At least one other seller defects, or buyer defects.* Then resolution does not occur regardless of $S_i$'s action:

$$u_{S_i}(\ldots, \mathrm{C}_i, \ldots) = u_{S_i}(\ldots, \mathrm{D}_i, \ldots) = -2G_i. \tag{23}$$

Payoffs are equal.

Combining both cases, C weakly dominates D for $S_i$, with strict inequality in Case 1. Since the choice of $i$ was arbitrary, this holds for all $n$ sellers. The buyer's dominance follows from Theorem 4.3(a), generalized: the buyer's payoff under cooperation ($-\sum P_i$) strictly exceeds the payoff under defection ($-2\sum P_i$) when all sellers cooperate.

Iterated elimination: in the first round, D is eliminated for all players (weakly dominated). The unique surviving profile is $(\mathrm{C}, \ldots, \mathrm{C})$. □

*Remark* 5.4 (DAG Independence). The process tree in Theorem 5.1 is stated as a chain for expositional clarity, but the Nash equilibrium result requires no sequential structure. Each order $o_i$ is an independently bonded bilateral commitment: the seller of $o_i$ bonds $2G_i$ and the buyer bonds $2P_i$ regardless of how many other orders share the process, what fan-in or fan-out topology the process exhibits, or how deeply nested within the DAG $o_i$ sits. The cooperation dominance argument of Theorem 5.3 applies to each order edge independently; it does not traverse the DAG and is not affected by the DAG's shape. The formal extension of Theorem 5.3 to arbitrary DAG topologies therefore adds no new insight: the equilibrium holds at every edge by construction of the bonding rule, not by induction over a particular graph topology. This is a deliberate design choice: bonding requirements are per-order primitives; the DAG is a composition structure that the mechanism neither reads nor enforces. The two concerns are orthogonal by design.

*Remark* 5.5 (Strength of the $N$-Party Result). Note that the seller's dominance condition, $P_i + 2G_i > 0$, holds with a margin that *grows* with chain depth (since $G_i = \sum_{j \le i} P_j$ is monotonic). The cooperation-defection payoff gap for $S_i$ is:

$$\Delta_i = P_i - (-2G_i) = P_i + 2G_i = P_i + 2\sum_{j=1}^{i} P_j \ge 3P_i. \tag{24}$$

The gap is at least $3P_i$ (equality only at the root where $G_i = P_i$) and grows with every upstream payment. This is the formal basis of the "geometric coordination pressure" in Theorem 5.6.

**Corollary 5.6** (Coordination Pressure Grows With Depth)**.** *The risk-to-reward ratio for seller $S_i$ is:*

$$\rho_i = \frac{C_{s,i}}{P_i} = \frac{2G_i}{P_i} = \frac{2\sum_{j=1}^{i} P_j}{P_i}. \tag{25}$$

*For equal payments ($P_j = p$ for all $j$), $\rho_i = 2i$, growing linearly with chain depth. For value chains where early stages contribute disproportionately ($P_1 \gg P_n$, with $P_n$ bounded below), $\rho_n$ scales as $2G_n/P_n$ and grows faster than linearly in $n$ relative to the late-stage contribution. The qualitative point is that the deeper participant absorbs the cumulative upstream risk; the precise growth rate depends on the shape of the payment schedule.*

**Example 5.7.** Alice orders food (\$10) from Bob, who needs delivery (\$2) from Charlie:

| Party | Role | $P_i$ | $G_i$ | $C_{s,i}$ | $\rho_i$ |
|---|---|---|---|---|---|
| Bob | Seller 1 | \$10 | \$10 | \$20 | 2.0 |
| Charlie | Seller 2 | \$2 | \$12 | \$24 | 12.0 |

Charlie risks \$24 to earn \$2 ($\rho = 12$). Bob risks \$20 to earn \$10 ($\rho = 2$). The deeper participant faces amplified consequences.

## 5.2 Self-Enforcing DAG Topology

Process trees can express arbitrary directed acyclic graph (DAG) topologies. A natural question: must the contract validate parent–child relationships on-chain? It need not. The bonding rule itself makes honest reporting of cumulative value the dominant strategy whenever the probability of non-resolution is positive.

**Theorem 5.8** (Self-Enforcing DAG Honesty)**.** *In any game where seller $S_i$ chooses a reported cumulative value $G'_i \geq G_i$ (with $G_i$ being the true value enforced by the contract's accumulator), truth-telling ($G'_i = G_i$) weakly dominates all alternatives. The dominance is strict whenever the probability of non-resolution is positive.*

*Proof.* The monotonic accumulator (Theorem 3.3) enforces $G'_i \geq G_i := \Pi.G_{\text{prev}} + P_i$: any report $G'_i < G_i$ causes the `commit` operation to revert (see Theorems 3.2 and 3.3, the `expectedCumulativeValue` check), so we restrict attention to $G'_i \geq G_i$.

Let $q \in [0, 1]$ be the probability that the buyer resolves the process (a function of all sellers' performance, exogenous to $S_i$'s reporting decision). Define $S_i$'s expected payoff as a function of the report $G'_i$:

$$\mathbb{E}[u_{S_i}(G'_i)] = q \cdot (\underbrace{2G'_i + P_i}_{\text{payout}} - \underbrace{2G'_i}_{\text{bond}}) + (1-q) \cdot (-2G'_i) = q \cdot P_i - (1-q) \cdot 2G'_i. \tag{26}$$

The derivative with respect to $G'_i$ is:

$$\frac{\partial}{\partial G'_i} \mathbb{E}[u_{S_i}] = -2(1-q). \tag{27}$$

For $q < 1$ (non-zero probability of non-resolution), this derivative is strictly negative: expected utility is strictly decreasing in the reported cumulative value. The optimum is therefore at the smallest feasible value, $G'_i = G_i$.

For $q = 1$ (certain resolution), the derivative is zero and the seller is indifferent—overstating wastes capital but does not affect the net payoff. Truth-telling remains weakly optimal.

Since $q < 1$ in any realistic setting (the seller cannot guarantee other sellers' cooperation), truth-telling strictly dominates overstating. □

*Remark* 5.9 (Why no on-chain DAG validation is needed). The theorem licenses a deliberate omission in the kernel: there is no on-chain enforcement of parent–child relationships in the process tree. The mechanism reads only the cumulative-value accumulator and the per-order bond; the DAG itself is reconstructed off-chain from event logs. Misreporting upstream lineage cannot profit a seller because the bonding rule already makes overstating $G_i$ strictly dominated.

### 5.3 Atomic Resolution and Seller Coordination

**Definition 5.10** (Atomic Resolution)**.** Resolution is *atomic*: the buyer must resolve all active orders in a process simultaneously, or none. There is no partial resolution.

**Proposition 5.11** (Endogenous Coordination Pressure)**.** *The $N$-party bonding game under atomic resolution induces a* weakest-link public goods game *among sellers, where each seller's payout depends on universal cooperation.*

*Proof.* Consider the sellers' subgame (fixing the buyer's strategy at: resolve iff all sellers cooperate). Each $S_i$ has payoff $u_{S_i}$ as in (19).

Define the *externality* imposed by $S_k$'s defection on $S_i$ ($i \neq k$):

$$\varepsilon_{k\to i} = u_{S_i}(\text{all C}) - u_{S_i}(S_k \text{ defects}) = P_i - (-2G_i) = P_i + 2G_i. \tag{28}$$

This is strictly positive for all $i, k$ with $i \neq k$. Every defection imposes a loss on every other seller. The total externality of $S_k$'s defection is:

$$\mathcal{E}_k = \sum_{i\neq k} \varepsilon_{k\to i} = \sum_{i\neq k}(P_i + 2G_i). \tag{29}$$

This structure satisfies the defining properties of a *weakest-link game* [Hirshleifer, 1983]:

1. The group outcome depends on the minimum individual effort (one defection collapses all payoffs to the non-cooperative level).
2. The cooperative outcome $(\text{C}, \ldots, \text{C})$ is the unique Pareto-optimal Nash equilibrium of the subgame.
3. The game exhibits *strategic complementarity*: $S_i$'s marginal return to cooperation is highest when all others cooperate.

In a repeated-interaction setting, this externality structure creates endogenous peer monitoring: $S_i$ has a direct economic interest of magnitude $P_i + 2G_i$ in ensuring $S_k$ cooperates. This incentivizes information acquisition about co-sellers, pressure application on underperformers, and reputation-based exclusion of known defectors—all without explicit communication protocols or governance mechanisms. □

*Remark* 5.12 (Distinction from the experimental weakest-link literature). The experimental weakest-link literature, originating with Van Huyck et al. [1990] (Van Huyck, Battalio and Beil, *AER* 1990), finds that weakest-link games empirically produce coordination *failure* in groups larger than two: players converge to the Pareto-inferior all-defect equilibrium via pessimistic beliefs about co-players. This might appear to threaten the result above. It does not. The classical experimental result arises when cooperation is *collectively* preferred but not *individually* dominant — a setting where pessimistic beliefs about others are sufficient to rationalise defection. In the bonded commitment mechanism, defection is individually dominated regardless of what others do (Case 2 of the proof of Theorem 5.3: even when all others defect, $S_i$ incurs $-2G_i$ whether it cooperates or defects, so there is no strategic incentive to defect). The cooperation pressure identified in Theorem 5.11 therefore arises in a setting where coordination failure through pessimistic beliefs cannot occur: each seller's dominant strategy is already to cooperate, and the weakest-link structure adds the externality characterisation on top of that result, not as a substitute for it.

### 5.4 Seller Non-Performance and Externality Handling

A natural question in any multi-party coordination mechanism is how negative externalities from one participant's non-performance are absorbed by the rest of the process. The bonded commitment mechanism handles this through the process structure itself, without requiring additional governance or protocol modifications.

**Seller substitution.** If seller $S_i$ fails to perform, the buyer holds a non-resolved process in which $S_i$'s bond is locked but delivery has not occurred. Before calling `resolveProcess`, the buyer may commit a replacement order with a new seller $S_i'$ at the same or renegotiated payment. $S_i'$ bonds against the current cumulative value, enters the process under the standard bonding rule, and delivers. When the buyer resolves, $S_i'$ receives full settlement. $S_i$, having failed to perform, forfeits their bond under the bonding equilibrium. The externality of $S_i$'s non-performance is therefore absorbed by $S_i$'s own capital loss, not distributed to co-sellers: other sellers' payouts are unaffected.

**External risk products.** Where seller substitution is impractical—latency-sensitive processes, highly specialised roles, or cases where a replacement cannot be sourced—standard risk instruments such as performance guarantees, surety bonds, and insurance can layer over the bonded process without modifying the kernel. These instruments hold independently bonded positions on the seller's behalf; they do not call `commit` or `resolveProcess`, hold no kernel state, and leave the bonding equilibrium intact. The mechanism is therefore not the only layer of externality protection available to parties; it is the settlement primitive over which such risk products compose.

## 6 Reduction from Microfinance Mechanisms

The coordination pressure of Theorem 5.11 is operationally analogous to the peer-enforcement dynamics documented in the microfinance literature on group lending—most prominently the Grameen Bank [Yunus, 1999] and its formalizations by Ghatak [1999] and Stiglitz [1990]. The analogy is frequently drawn informally, but the precise relationship between the bonded commitment mechanism and the microfinance models has not, to our knowledge, been stated formally. We make it explicit here because it identifies what FIGARO actually contributes relative to the prior literature on peer-enforced cooperation.

**The microfinance mechanisms.** Two canonical models frame group lending. Ghatak [1999] proves *peer selection*: under joint-liability contracting, borrowers with private type information sort assortatively, attenuating adverse selection. The result requires a heterogeneous population, local information among borrowers, and voluntary group formation. Stiglitz [1990] proves *peer monitoring*: joint liability combined with repeated social interaction lets group members observe and punish co-borrower deviation, attenuating moral hazard. The result requires repeated interaction, observability of co-borrower actions, and an exogenous punishment technology (social sanction, denial of future credit). In both cases the peer-pressure outcome is an *emergent* property of a contract (joint liability) embedded in a social substrate (repeated interaction, local information). The contract alone is insufficient; the substrate does enforcement work.

**Proposition 6.1** (Assumption Reduction on Cooperation Pressure)**.** *The* cooperation-pressure component *of the equilibrium common to Ghatak [1999] and Stiglitz [1990]—cooperative equilibrium selection driven by co-participant externalities—is obtained by the bonded commitment mechanism under strictly weaker assumptions. Specifically, FIGARO requires none of the following: (i) repeated interaction, (ii) local information among sellers, (iii) a punishment technology exogenous to the contract, nor (iv) joint-liability contracting.*

*Proof.* Theorem 5.3 establishes cooperation as the unique profile surviving iterated elimination of weakly dominated strategies in a one-shot game. The argument uses only the on-chain payoff function and makes no appeal to continuation value, trigger strategies, or folk-theorem constructions; (i) is not required.

The externality $\varepsilon_{k\to i} = P_i + 2G_i$ (Theorem 5.11) is computed from quantities enforced by the settlement layer's monotonic accumulator, independent of any seller's information about co-sellers' types, actions, or identities; (ii) is not required.

The punishment for a co-seller's defection is the loss of $S_i$'s own bond $C_{s,i} = 2G_i$, imposed directly by the contract through non-resolution. No external punishment technology—social sanction, reputational downgrade, or denial of future credit—is invoked; (iii) is not required.

Finally, each $C_{s,i}$ is posted individually by $S_i$ against $S_i$'s own cumulative-value snapshot, not against the group's aggregate obligation. Coupling across sellers arises from the buyer's atomic-resolution constraint (Theorem 5.10), not from the bond structure; (iv) is not required. □

**What is *not* reproduced.** The proposition is scoped deliberately. Group-lending models derive two distinct equilibrium effects, and FIGARO reproduces only the first.

- **Cooperation under externality coupling (reproduced).** Once a group is formed and joint liability is in force, each member has incentive to perform because a co-member's failure imposes direct loss. The bonded mechanism reproduces this effect under strictly weaker assumptions as shown above.
- **Peer selection under private type information (not reproduced).** Ghatak [1999] shows that joint-liability contracts induce *assortative matching* among borrowers of different unobservable risk types; this is an adverse-selection result and requires both a heterogeneous type space and a group formation stage. FIGARO has no type-screening mechanism and no group-formation stage that selects on unobservable seller quality. Adverse-selection screening must be obtained by some mechanism outside the bonded commitment (e.g., attested credentials, operator registries).
- **Peer monitoring under principal information asymmetry (not directly reproduced).** Stiglitz [1990] treats a setting where deliverable quality is unobservable to the principal (lender) but observable to peer borrowers. The equilibrium leverages the principal–peer information asymmetry. FIGARO's bonded commitment assumes the buyer can determine whether to resolve; it sidesteps rather than solves the buyer-side moral hazard. When the buyer's performance observation is itself noisy or delayed, the perfect-monitoring assumption in Theorem 5.2 becomes load-bearing and the cooperation-pressure reproduction depends on how that noise is handled at the mechanism layer.

The honest summary of Theorem 6.1 is therefore that it reproduces one of the two classical group-lending equilibrium effects, under weaker assumptions, for a correspondingly narrower conclusion. The other effect — assortative screening on unobservable types — requires additional mechanism above the bonded kernel.

**What this repositions.** The reduction does not correct the microfinance theorems—they are sound under their assumptions—but it reframes them. Repeated interaction and local information were necessary for group lending because the enforcement primitive available in that setting was reputation within a social network. Once the settlement layer admits bilaterally signed, individually collateralized commitments with atomic resolution, the same mechanism outcome obtains among mutual strangers, in a single shot, with no social substrate. Repeated interaction and local information are thus not necessary conditions for endogenous peer monitoring; they are features of one particular settlement environment in which the mechanism was historically deployed. The weakest-link subgame (Theorem 5.11), the geometric risk amplification (Theorem 5.6), and the one-shot dominance argument (Theorem 5.3) together give a purely mechanism-level construction of an effect the microfinance literature attributes to contract design combined with social embedding.

## 7 Conclusion and Open Questions

We have presented FIGARO, a settlement primitive that achieves self-enforcing multi-party coordination through two composing mechanisms: *asymmetric bonding*, which produces the bilateral Nash equilibrium and scales it to $N$-party process trees via progressive collateralization; and *buyer dominance with atomic resolution*, which enforces inter-seller coordination on the already-scaled mesh. The kernel is minimal—no timeout, no governance, no escape hatches—and the game-theoretic properties are strong: cooperation is weakly dominant at every position in an $N$-party process tree, and the cooperative profile is the unique outcome surviving iterated elimination of weakly dominated strategies. The mechanism reproduces the peer-enforcement effect of joint-liability microfinance under strictly weaker assumptions, with no appeal to repeated interaction, local information, or social sanction.

**Implementation and verification.** A reference implementation as a Solidity smart contract, together with the formal-verification stack (TLA$^+$, symbolic execution, property-based fuzzing, specification checking) and the threat model that accompanies it, is presented in the companion implementation paper. That paper also develops the *coordinator pattern*: sufficient conditions under which an external mechanism composed with the kernel preserves the bonding equilibrium, and the corresponding three-layer enforcement architecture (economic / coordination / evidence).

**Organizational implications.** The implications of a fixed-cost enforcement primitive for the boundary of the firm in the sense of Coase [1937]—and specifically for the demotion of the firm from privileged organizational container to one pattern among many (alongside transaction-scoped assemblies, treasury communities, peer networks, and agent coalitions) when the subordination overlay is no longer enforcement-driven—are developed in the companion institutional-economics paper.

**Focal-point coordination among participants.** The companion FIG token paper develops FIG, a focal-point coordination token that operates without any kernel-level interaction: it does not call `commit` or `resolveProcess`, holds no kernel bonds, and modifies no kernel state, thereby preserving the bonding equilibrium trivially.

**The accounting byproduct.** A consequence of the conservation law (Theorem 3.4) worth flagging: the kernel is at all times in custodial balance, and a process is, in the precise accounting sense, a self-closing ledger period. The companion accounting paper develops the implications for the bookkeeping apparatus that grew up around double-entry [Pacioli, 1494] once that apparatus is made optional by protocol-level constitution of the records.

**Multi-currency scope exclusion.** The mechanism is defined for a single denomination per process. Mixing currencies within one process requires an external rate of substitution—oracle, DEX, or pre-agreed FX rate—each of which reintroduces a discretionary actor and breaks the same-unit comparability on which the $2\times$ Nash-stability result rests. This is an exclusion by design, not an open problem. Multi-token vendor workflows are handled at the wallet layer: a participant swaps into the process denomination before calling `commit`, so the kernel invariant is preserved trivially.

**Open theoretical question.** The relationship between the 2× multiplier and the opportunity cost of locked capital warrants closer treatment: the present analysis shows that $k = 2$ is sufficient at zero opportunity cost and remains sufficient for $rt > 0$, but does not characterize the rate regime under which a richer dynamic mechanism (e.g., interest-bearing bonds, time-varying multipliers) would strictly Pareto-dominate the static $k = 2$ design.

**A scope note on voluntary participation.** The equilibrium results of this paper hold under the assumption that both parties are legally and materially free to enter the bonding game — i.e., that entering is a choice and that capital sufficient to post the required bond is available. For participants whose only realistic income path runs through bonded assemblies, or who lack access to the bond-denomination currency, the "voluntary" framing is itself a variable whose treatment lies outside mechanism design. The distributional and labor-law consequences of this scope are developed in the companion institutional-economics and labor-law papers; the present paper establishes equilibrium properties conditional on voluntary participation, not the conditions under which participation is voluntary in any substantive sense.


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